ورقة بحثية أصلية — الترجمة الأكاديمية الحرفية الكاملة لجميع الأقسام والفقرات (10 صفحة)

تقييم التعرف على إيماءات اليد المعتمد على التآزر العضلي تحت تغيرات قوى الإمساك

Evaluation of Synergy-Based Hand Gesture Recognition Under Force Variation

مينغ تنغ، تشن هوانغ، وتشوان شين

الترجمة الأكاديمية الكاملة لرسالة الماجستير في هندسة الميكاترونيات — جامعة عين شمس

Evaluation of Synergy-Based Hand Gesture Recognition Method Against Force Variation

الملخص- الخصائص غير الثابتة للسطح

Abstract— The non-stationary characteristics of surface

تخطيط كهربية العضل (sEMG) والتغيرات الضارة المحتملة في ظروف العالم الحقيقي تجعل الأمر لا يزال يمثل تحديًا مفتوحًا لتحقيق التحكم القوي في كهربية العضل (MEC) للأطراف الاصطناعية متعددة الوظائف. يعد مستوى تقلص العضلات المتغير أحد العوائق التي قد تؤدي إلى انخفاض أداء MEC. في هذه الدراسة، اقترحنا طريقة للتعرف على نية القوة الثابتة بناءً على تحليل تآزر العضلات (MSA) في تحديد ثلاثة مستويات قوة محددة ذاتيًا (منخفضة ومتوسطة وعالية). على وجه التحديد، تم اختيار خوارزمية تحليل المصفوفة السريعة المستندة إلى المربعات الصغرى غير السلبية المقيدة بالتناوب (NMF/ANLS) لاستخراج أوجه التآزر الخاصة بالمهمة المرتبطة بكل من إيماءات اليد الست في مرحلة التدريب؛ بينما بالنسبة لعينات الاختبار، استخدمنا طريقة المربع غير السالب (NNLS) لتقدير الأوامر العصبية لتصنيف الحركة. وتمت مقارنة أداء الطريقة المقترحة مع طريقة التعرف على الأنماط التقليدية (PR) التي تتكون من مصنف LDA (تحليل التمييز الخطي) وميزات تمثيلية في ثلاثة سيناريوهات تقييم غير متصلة بالإنترنت. كشفت الاختبارات الإحصائية التي أجريت على عشرة أشخاص أصحاء عن عدم وجود فرق كبير في دقة مستوى القوة الداخلية (ع = 0.353) ومستوى القوة المتعددة (ع = 0.695)؛ ولكن كان أداء الطريقة القائمة على التآزر أفضل بكثير من الطريقة التقليدية القائمة على العلاقات العامة في ظل ظروف مستوى القوة المشتركة (P <0.05). ولوحظت نتائج مماثلة لتسع أشخاص مبتوري الأطراف على الرغم من وجود انخفاض في دقة التصنيف. كانت هذه الدراسة هي الأولى التي أظهرت حاليًا القوة والقوة التنبؤية للتآزرات الخاصة بالمهام في ظل مستويات القوة المتغيرة واستكشفت إمكاناتها للتعرف بشكل موثوق على النوايا مقابل تباين القوة. على الرغم من أن الأداء عبر الإنترنت لم يتم إثباته بعد، فإن الطريقة المقترحة تتميز بإجراءات تدريب بسيطة وكفاءة حسابية مقبولة، والتي من المحتمل أن توفر بديلاً.

Manuscript received January 31, 2021; revised June 6, 2021 and September 26, 2021; accepted October 24, 2021. Date of publication November 2, 2021; date of current version November 11, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 51775415, in part by the Key Research and Devel- opment Program of Shaanxi Province of China under Grant 2018ZDCXL- GY-06-01, and in part by the Special Guidance Funds for the Construction of World-Class Universities/Disciplines and the Characteristic Development in Central Universities under Grant PY3A071. (Corresponding author: Guanghua Xu.)

شمل هذا العمل موضوعات بشرية أو حيوانات في أبحاثه. تم منح الموافقة على جميع الإجراءات والبروتوكولات الأخلاقية والتجريبية من قبل لجنة الأخلاقيات بجامعة سيدني للتكنولوجيا وجامعة بليموث.

المؤلفون يعملون في كلية الهندسة الميكانيكية، معهد الدراسات متعددة التخصصات للهندسة والطب، جامعة شيان جياوتونغ، شيان 710049، الصين (البريد الإلكتروني: ghxu@xjtu.edu.cn).

Digital Object Identifier 10.1109/TNSRE.2021.3124744

للتحكم القوي في كهربية العضل

تشيتشنغ تينغ، غوانغهوا شو، عضو، IEEE، رينجهاو ليانغ، مين لي، عضو، IEEE، وسيكونغ تشانغ

approach for the developmentof clinicallyviable prostheses .and rehabilitation robots driven by sEMG

مصطلحات الفهرس - sEMG، التحكم الكهربائي العضلي، الأطراف الاصطناعية متعددة الوظائف، مستوى تقلص العضلات، تآزر العضلات.

I. INTRODUCTION

I. INTRODUCTION

OWERED myoelectric prostheses can, to some extent, help amputees regain the lost function of their residual limbs and rebuild confidence in daily life. Despite the fact that the mechanical design of state-of-the-art robotic hands enables vivid movements imitation of human hands [1], these dexterous anthropomorphic hands are beyond direct appli- cation in myoelectric prostheses due to the incommensurate performance .of available human-machine-interface (HMI)

Over the past decades, pattern recognition (PR) based myoelectric classification scheme has been widely explored for multifunctional prostheses control [2], [3]. Such scheme employs multichannel EMG signals and provides intuitive control

with more active degree of freedoms (DOFs). Despite a large amount of promising results (with the average accuracy above 95%) reported in controlled laboratory environments, there exists a noticeable disparity between ideal and real- world conditions. It has been confirmed that the reliability of myoelectric control was highly dependent on physical and physiological constraints, including fixed electrode location on specified muscles, specific limb orientation, steady muscle contraction level, avoidance of muscle fatigue, and so on. These favorable conditions, however, are not likely to meet in a practical context for prostheses control. It is for these reasons that PR-based control scheme meets its Waterloo in clinical translation. To our knowledge, only one spin-off company [4] offers a classification-based controller as an add-on for various prostheses. Consequently, the rejection rate of multifunctional myoelectric prostheses and the use of non-functional cosmetic prostheses remains stay at a high level [5], [6].

Researchers have made multipronged efforts to improve the performance of PR-based control scheme in the presence of variant muscle contraction levels. Tolooshams and Jiang [7] and Pradhan et al. [8] proposed a signal pre-processing method using frequency division technique (FDT) for online myo- electric PR classification. The results indicated that FDT had an advantage over traditional methods in the tested real-time myoelectric control tasks. As for the classifiers, Li et al. [9]

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.

اقترح استراتيجية تصنيف متوازية تتكون من ثلاثة مصنفات تم تدريبها باستخدام البيانات التي تم الحصول عليها عند مستويات قوة مختلفة، وصممت لائحة لتصنيف إشارات الإدخال إلى مصنفات مقابلة من أجل علاقات عامة قوية. معين وآخرون. [10] استخدم نموذجًا للحوسبة فائقة الأبعاد مستوحى من الدماغ لبناء نماذج تصنيف قوية لفرض الاختلافات وقادرة على التعرف على مستويات الانكماش المتعددة. اقترح Powar و[11] Chemmangat طريقة التزييف الزمني الديناميكي (DTW) لتقليل تأثير اختلافات القوة على EMG PR، وحققا زيادة كبيرة بنسبة 10٪ في دقة التصنيف. اقترح الباحثون أيضًا تقنية استخلاص الميزات القوية (FE) ضد تباين القوة لنظام EMG PR، وتم الإبلاغ عن بعض النتائج المشجعة [12]-[15].

قد يكمن أحد أوجه القصور الرئيسية في نظام التصنيف القائم على العلاقات العامة في عدم كفاءته في الاستفادة الكاملة من المعلومات الفيزيولوجية العصبية الأساسية المخفية في بيانات sEMG. بمعنى آخر، لا يمكن دمج المعلومات المهمة المرتبطة بعمليات التحكم العصبي لتوليد أنماط تنشيط العضلات التي يمكن ملاحظتها (MAP) بشكل صريح في النظام القائم على العلاقات العامة [16]. على الرغم من أن تضمين جميع البيانات عند مستويات تقلص العضلات المختلفة في إجراء التدريب قد يخفف من هذه الآثار الضارة، إلا أن زيادة وقت التدريب وتحسين الأداء المحدود لا يزال يمثل مشكلة للترجمة السريرية لهذه التكنولوجيا.

تم اقتراح التآزر العضلي (MS) باعتباره لبنات بناء يمكن أن تبسط بناء السلوكيات الحركية المعقدة. تشير هذه النظرية إلى أن الجهاز العصبي المركزي (CNS) قد لا يقوم بتنشيط العضلات الفردية بشكل مباشر واحدة تلو الأخرى لتحقيق التحكم الحركي. بدلا من ذلك، يتم تنشيط مجموعة من العضلات في وقت واحد عن طريق الأوامر العصبية الزمنية من الجهاز العصبي المركزي لأداء حركات DOF متعددة بطريقة منسقة [17]. على الرغم من أن وجود مرض التصلب العصبي المتعدد لا يزال سؤالًا مفتوحًا، فقد تم تطوير العديد من أنظمة HMI المتقدمة باستخدام إطار عمل MS. جيانغ وآخرون. [18]، [19] تم استخراج حركيات المعصم من EMG متعدد القنوات باستخدام خوارزمية تحليل المصفوفة غير السلبية الحكيمة DOF لتحقيق التحكم التناسبي والمتزامن في اثنين أو ثلاثة DOFs. ما وآخرون. [20] اقترح مخطط تحكم يعمل على تسهيل المدخلات القائمة على التآزر، وبالتالي تثبيت التحكم في الحركة. ايسون وآخرون. [21]، [22] أظهر تحكمًا متزامنًا ومتناسبًا للواجهة الكهربية العضلية DOF-4 باستخدام مدخلات مستوحاة من تآزر العضلات من EMG عالي الكثافة. ومع ذلك، قدمت كل هذه الدراسات سيطرة موثوقة على حساب DOF محدود. استكشف الباحثون أيضًا دور مرض التصلب العصبي المتعدد في تصنيف حركة DOF المتعددة. ومع ذلك، فقد استخدموا إما تآزر العضلات المستخرجة أو معاملات التنشيط كمتجه للميزات لتدريب المصنف على تمييز المهام، والذي كان في الأساس لا يزال نظام علاقات عامة تقليدي ولم يتم التحقيق في القوة التنبؤية لمرض التصلب العصبي المتعدد بشكل كامل [23] - [26]. تحقق Ajiboye و[27] Weir من الإطار التنبؤي لمرض التصلب العصبي المتعدد عن طريق استخلاص ثمانية أوجه تآزر من أحد عشر وضعًا للسماح بتنبؤ 90٪ على الأقل بأنماط EMG لجميع المواقف الـ 33، والتي تلقي الضوء على تصنيف الحركة القائم على التآزر. رسول وآخرون. [16] ثم قدم صيغة جديدة تستخدم التآزر الخاص بالمهمة وتمثيل مساحة الحالة للإشارات العصبية لتمييز حركات اليد الست. توسيع عمل رسول، تشانغ وآخرون. [28] تم بناء مخطط التحكم المتزامن بواسطة

باستخدام مُصنف واحد قائم على التآزر لكل إيماءة من الإصبع. ومع ذلك، لم يبحث أي من الأعمال المذكورة أعلاه في دور مرض التصلب العصبي المتعدد في تطوير مخطط تصنيف القوة الثابتة. باتيل وآخرون. [29] اقترح طريقة تصنيف ضد تباين القوة مستوحاة من تنسيق العضلات. على الرغم من أنها غنية بالمعلومات، إلا أنها فشلت في استخدام نموذج تآزر جيد البناء لتحليل تآزر العضلات. تابوري وآخرون. [30] قام بتقييم متانة إعادة بناء فريق الإدارة البيئية عبر مرض التصلب العصبي المتعدد وتكرار معلمات مرض التصلب العصبي المتعدد كمؤشرات فسيولوجية عصبية محتملة. دينغ وآخرون. [31] وجينج وآخرون. [32] أوضح تعديل مرض التصلب العصبي المتعدد لحركات الساعد المتعددة تحت مستويات القوة المتغيرة ومواضع الذراع. ومع ذلك، فإنهم لم يقترحوا خطة تصنيف تستخدم مرض التصلب العصبي المتعدد لتحقيق تصنيف قوي للحركة.

In the present study, we concurrently demonstrated the effect of variant muscle contraction levels on both the constitution of estimated MS and the performance of a synergy-based classification method. Firstly, a fast matrix factorization algorithm based on alternating non-negativity constrained least squares (NMF/ANLS) was used to extract task-specific synergies for six hand gestures. Then the structure similarity and predictive

power of estimated synergies were investigated in the setting of three self-defined muscle con- traction levels. Following that, a novel classification scheme against force variation based on muscle synergy analysis was proposed accordingly. Finally, the performance of proposed algorithm was evaluated and compared with conventional PR-based method consisting of linear discrimination analy- sis (LDA) classifier and representative features reported in literature. The outcome of this study would benefit the design of clinically viable prosthetic hand.

ثانيا. المواد والأساليب
II. MATERIALS AND METHODS

A. sEMG Data Collection and Preprocessing

نحن نستخدم بيانات فريق الإدارة البيئية من مجموعتين من البيانات المتاحة للجمهور لإثبات فعالية نظام التصنيف المقترح. نظرًا لطول الورقة، تم استخدام مجموعة البيانات الأولى فقط لتوضيح المتانة والقوة التنبؤية لتآزر العضلات. يتم تقديم وصف موجز لمجموعتي البيانات هاتين هنا من أجل السلامة الهيكلية لهذه الورقة.

1) مجموعة البيانات القادرة على العمل: مجموعة البيانات الأولى من خوشابا وآخرون. [33] يتضمن إشارات EMG مسجلة من عشرة أشخاص أصحاء. تتراوح أعمارهم بين 20 إلى 30 عامًا، ويبلغ قطر الساعد 26.6 ± 2.4 سم. لم يكن لدى المشاركين أي خبرة سابقة في استخدام HMI الذي يتحكم فيه فريق EMG. وقد تم منحهم جلسة إحاطة مدتها 20 دقيقة للتعرف على البروتوكول التجريبي. تمت الموافقة على جميع الإجراءات من قبل لجنة أخلاقيات البحث في جامعة التكنولوجيا في سيدني.

During the experiment, subjects sat comfortably in front of a standard computer screen. Instructions appeared clearly on the computer screen using an image of the desired move- ment along with a text caption. The subjects performed six movements including hand close (HC), hand open (HO), wrist extension (WE), wrist flexion (WF), wrist ulnar deviation (WU), and wrist radial deviation (WR). Each of these six movements was repeated three times at three self-defined muscle contraction levels, i.e., low (L), medium (M), and high (H). A virtual instrument implemented in LABVIEW

was used for signal acquisition (National Instruments, USA) and display, which helps the subjects produce the needed force level. The subjects were asked to hold each gesture for a duration of five seconds in each trial. There is a 10 s rest between trials to avoid muscle fatigue. The sEMG signals were collected at the sampling rate of 4000 Hz, with six electrodes (Bagnoli desktop EMG system, Delsys Inc., USA) equally spaced across the circumference of the forearm for each subject. This dataset also contains wrist accelerometer data, which was not included in the present study. For more information about the metadata, please refer to [33].

Amputee Dataset: The second dataset from Al-Timemy et (2 al. [12] consists of EMG signals recorded from nine transradial amputees (seven traumatic and two congenital). All subjects missed their left hand except one with right hand amputation. The first seven subjects had never used myoelectric prostheses while the remaining two used one for a certain time. The local ethical committee at the School of Computing and Mathematics, Plymouth University approved this research.

تم أخذ ست حركات في الاعتبار في التجربة، بما في ذلك ثني الإبهام، وثني السبابة، والقبضة الدقيقة، وقبضة الحامل ثلاثي الأرجل، وقبضة الخطاف، والقبضة الكروية. تم تنفيذ كل حركة على ثلاثة مستويات محددة لتقلص العضلات، أي منخفضة ومتوسطة وعالية. لكل مستوى قوة، تم تسجيل خمس إلى ثماني تجارب لكل مبتور حيث تم إجراء كل تجربة لمدة 8-12 ثانية. استخدم مبتوري الأطراف أيديهم السليمة لمساعدتهم على تخيل الحركة المطلوبة بالقوة المطلوبة. بالإضافة إلى ذلك، استخدموا ردود الفعل المرئية من شاشة LABVIEW لمعرفة كيف يتغير سعة EMG مع القوة، وهو ما كان مفيدًا لهم لإنتاج مستوى القوة المطلوب. تم جمع إشارات EMG بمعدل أخذ عينات قدره 2000 هرتز، مع وضع ثمانية أزواج من أقطاب Ag / AgCl (Tyco للرعاية الصحية، ألمانيا) حول الجذع لكل مبتور الأطراف. تم استخدام بيانات من التجارب الثلاث الأولى في هذه الدراسة. للحصول على وصف تفصيلي لمجموعة البيانات، يرجى الرجوع إلى [12].

Preprocessing: The recorded raw sEMG signals from the (3 datasets for each movement were then band-pass filtered (4th order Butterworth filter, 20-450 Hz), full-wave rectified, and low-pass filtered (4th order Butterworth filter, 20 Hz) to obtain the envelope of the sEMG signals.

ب. نموذج التآزر العضلي
B. Muscle Synergy Model

The time-invariant muscle synergy model indicates that the high-dimensional MAP can be represented as linear combina- tion of low-dimensional muscle synergies that are activated by corresponding activation coefficients [17]

مرحبا (ر) واي، (1)

= m (t)

where m(t) is a vector of MAP for a group of muscles at time t, each wi, namely the muscle synergy, is a vector representing a fixed weighting of activation within the mus- cles, each hi(t) is the time-varying combination coefficients hypothesized as the neural commands for the activation of wi, k is the number of muscle synergies. Equation (1) can be

مكتوبة كنموذج مصفوفة فيما يتعلق بحالات زمنية n على نطاق زمني محدد:

مم × ن = وم × ك هك × ن

ج1 ج2

....

IIII

IIII، (ك < م) (2)

w1 w2 · · · wk =

... هونج كونج

where M is the MAP matrix of m muscles, W is the muscle synergy matrix, H is the activation coefficient matrix. Please note that M is known whereas W and H is unknown in this case.

ج. تقريب أنماط تنشيط العضلات

C. Muscle Activation Patterns Approximation

The recorded sEMG data was processed and divided into several blocks, with each block corresponding to a specific hand gesture. Data from each block were then segmented into 250 ms analysis window with an increment of 30 ms, and features were extracted from each analysis window to consti- tute the feature matrix M for each hand gesture. This matrix was referred to as the MAP matrix. The resulting samples were then normalized for each subject and each muscle to the maximum activation in that muscle over all trials of all tasks.

The most commonly used method for MAP approxima- tion in muscle synergy analysis (MSA) is the EMG linear envelopes, researchers also used typical time domain (TD) features, i.e., root mean square (RMS) and mean absolute value (MAV) to produce MAP matrix. Ma et al. [20] evaluated the performance of a set of TD features in MSA by comparing the goodness of reconstruction using extracted MS, and find that the best three features are MAV, RMS and waveform length (WL). In this study, however, we put more focus on the influence of different features on classification accuracy and seeking to find the best MAP approximation method for synergy-based movement recognition. The features evaluated in this study were RMS, MAV, and a feature based on discrete Fourier transform and root (DFTR) [13].

D. Extraction of Muscle Synergies

D. Extraction of Muscle Synergies

Matrix Factorization Method: Muscle synergies can then be (1 estimated from the MAP matrix using factorization algo- rithms, i.e., non-negative matrix factorization (NMF) [34], independent component analysis (ICA) [35], and principle component analysis (PCA) [36]. NMF is widely used since it constrains the estimated synergies to be no-negative, and thus allowing for more physiological interpretation of the results (muscles activity and neural control signals are inherently no-negative). In contrast, The results of PCA and ICA are either orthogonal or statistically independent. We accordingly chose a NMF algorithm based on alternating non-negativity constrained least squares (NMF/ANLS) that showed fast overall convergence and generated more physically reliable solutions [37].

It should be noted that the NMF/ANLS algorithm usually generates multiple solutions for Equation (2) once the synergy number k is determined. For this reason, the algorithm was

تم تكرارها 10 مرات في هذه الدراسة لاستخراج مجموعة التآزر العضلي بأقل خطأ لإعادة بناء البيانات. استخدمنا معامل التحديد (R2) [38] كمعيار لقياس جودة إعادة الإعمار لبيانات sEMG الأصلية من خلال تآزر العضلات المقدرة ومعاملات التنشيط المقابلة:

R2 = 1 −‖M −W H‖2

(3) 2

م -ـم

حيث يشير ‖∗‖2 إلى معيار Frobenius التربيعي للمصفوفة، وـM هو متوسط مصفوفة MAP.

We also computed the variance accounted for (VAF), which is a similar criterion calculated on uncentered data:

V AF = 1 −‖M −W H‖2

M‖2 (4)‖

Determination of Synergy Number: The optimal number of (2 muscle synergies (dimensionality of neural commands from CNS) in the tested model is a hyper-parameter that should be determined before synergy extraction. In general, k is determined as the minimum number of muscle synergies able to explain the structural variation of the dataset, that is, addi- tional synergies only capture noise-generated variation [38]. We plotted the curve of R2 as functions of synergy number k (changing from 1 to 6) to depict the influence of synergy number on the reconstruction goodness of the original data. In this study, the optimal synergy number was determined at the point where R2 > 0.9, namely at least 90 percent of the variance in the data could be explained by the reduced dimension. To avoid data leakage, we only used one out of three EMG trials to separate the data used for determining the number of synergies from the data used for the accuracy testing. Reported for each movement performed by each subject under each force level were the optimal number for estimated muscle synergies and the corresponding R2.

هـ. متانة عمليات التآزر بين المهام المحددة

E. Robustness of Task-Specific Synergies

لكل حركة يقوم بها كل موضوع في كل مستوى من مستويات تقلص العضلات، تم استخراج مجموعة من التآزر العضلي من مصفوفة MAP المقابلة باستخدام خوارزمية NMF/ANLS. يعد الهيكل التآزري القوي لحركة معينة، أو خصوصية المهمة المرتبطة بالتآزر العضلي [39]، مهمًا في هذه الدراسة لأنه يزيد من إمكانية التصنيف الموثوق للحركة. ولتحقيق هذه الغاية، قمنا بتقييم متانة التآزر المقدّر الخاص

بالمهمة المحددة في ظل وجود ثلاثة مستويات قوة محددة ذاتيًا لكل موضوع.

Variation of Synergy Number: Synergy number has a direct (1 influence on the constitution of estimated muscle syner- gies, which stands a good chance of affecting the performance of synergy-based classification scheme proposed in Section II-G. In view of this, the optimal number of task-specific synergies in the presence of variant force levels was investi- gated herein for each able-bodied subject. And on top of that, we sought to find a generalized synergy number that would work for MSA under all tested conditions.

Robustness of Muscle Synergies: It is well known that the (2 MAP involved in a certain task are directly affected by

اختلاف مستويات تقلص العضلات. ولذلك، فمن الضروري التحقيق في متانة التآزر المقدرة ضد تباين القوة. لقد افترضنا أن تغيير مستوى تقلص العضلات ضمن حركة معينة لن يؤثر بشكل كبير على توازن الوزن ولكن فقط على معامل كل متجه تآزر. تم حساب متوسط جيب التمام للزاوية (ACA) بين كل زوج من متجهات التآزر كمعيار لقياس تشابهها.

i · wB

ACA = 1

أنا وا

، ط = 1، 2، . . . ، ك (5)

where · denotes dot product, wA

i and wB

i هو متجه التآزر i لأوجه التآزر الخاصة بالمهمة المقدرة تحت مستوى القوة A Bg على التوالي. وتتراوح النتيجة العددية للمعادلة (5) من صفر إلى واحد، حيث يشير الواحد إلى أعلى تشابه. في هذه الدراسة، تمثل القيمة العددية التي تزيد عن 0.8 تشابهًا كبيرًا بين تآزرين عضليين نظرًا لملاحظة نتائج تصنيف قابلة للمقارنة مع الطرق المرجعية في هذا الإعداد.

It should be noted that the synergy vectors in two compared muscle synergies may not perfectly match in order, this is due to the indetermination of NMF/ANLS algorithm. In light of this, we used a best matching algorithm to align the synergy pairs from two estimated synergy matrix. Specifically, the cosine similarity was calculated for all possible combination of synergy pairs between two estimated synergy matrices. The pair with the highest similarity was deemed a match in this study, and the process repeated until all synergy vectors in question were well aligned.

F. Predictive Power of Task-Specific Synergies

F. Predictive Power of Task-Specific Synergies

يتم تعريف القوة التنبؤية للتآزرات الخاصة بالمهمة هنا لوصف إعادة بناء البيانات الأصلية المرتبطة بمهمة معينة باستخدام التآزر العضلي المقدر من نفس المهمة. وهذا يعني أن مصفوفة التآزر W الخاصة بالمهمة المرتبطة بحركة واحدة تحت مستوى قوة واحد ظلت ثابتة لتقدير مصفوفة معاملات التنشيط H من مصفوفة MAP M داخل نفس الحركة تحت مستوى قوة آخر:

ح = ث + م (6)

حيث W+ هو المعكوس الزائف لأوجه التآزر المقدرة الخاصة بالمهمة المحددة. أشارت نتائج القسم III-A إلى أن معامل نواقل التآزر يتزايد مع اختلاف مستويات تقلص العضلات. للتخفيف من تأثير تباين القوة على التآزر العضلي المقدر، تم تطبيع مصفوفة التآزر W عبر القنوات قبل وضعها في المعادلة (6).

Then the derived activation coefficients matrix H were mul- tiplied with the previously estimated task-specific synergies W to reconstruct the corresponding MAP matrix M. The good- ness of reconstruction was measured by the VAF criterion. For comparison, task-specific synergies associated with a certain movement were also used to reconstruct MAP matrix of other different movements. Reported for each movement performed by each subject at each force level was the goodness of reconstruction measured by the VAF.

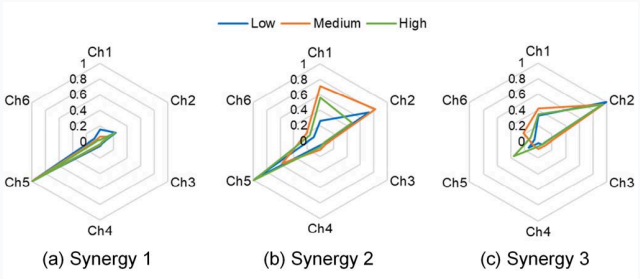


Fig. 1. Synergy-based classification scheme for movement recognition.

الشكل 1: .recognition

Fig. 1. Synergy-based classification scheme for movement recognition.

G. Movements Identification Based on Muscle

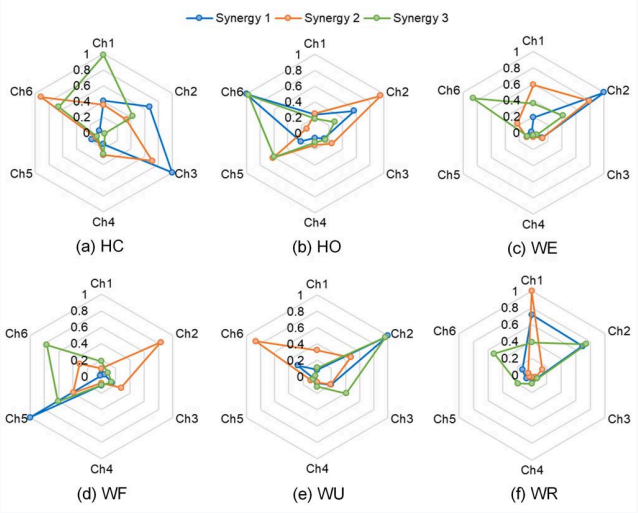
G. Movements Identification Based on Muscle

An overview of the proposed hand gesture recognition method based on muscle synergy analysis was illustrated in Fig. 1. The assumption here was that the task-specific synergies tended to reconstruct the sEMG data within the same hand gesture better than data associated with other different hand gestures. This assumption was verified in Section III-B.

The whole scheme mainly consisted of two stages: In the training stage, the task-specific synergies W N (N = 1, 2, . . . , 6) for each hand gesture was extracted from cor- responding MAP matrix MN using NMF/ANLS algorithm. These synergy matrixes were then provided one by one to reconstruct the testing data in the testing stage. For each MAP vector v extracted from the analysis window of testing data, the neural commands hN corresponding to each previously learned task-specific synergies were derived by using non- negative least square (NNLS) [40] method:

power spectrum descriptors (TDPSD) both achieved good performance against force variation when combined with linear discrimination analysis (LDA) classifier [12], [13]. We thus chose these two representative methods as the bench- mark for conventional PR system. We also evaluated the effect of different FE techniques on classification accuracy. Specifi- cally, the MSC method was evaluated using RMS, MAV, and DFTR, respectively. Whereas the LDA method was evaluated using DFTR, TDPSD, and TD feature set consisting of MAV, WL, zero crossings (ZC), and slope sign changes (SSC) [41]. Overall, six classification schemes, namely MSC + RMS, MSC + MAV, MSC + DFTR, LDA + TD, LDA + DFTR, LDA + TDPSD were considered in this study.

One-way analysis of variance (ANOVA) was utilized to compare the performance of different classification schemes. Separate ANOVA test was also applied on the MSC method to validate the effect of different training strategies on achieved



الشكل 2: Fig. 2. The R2 versus k plot for each hand gesture. The results were mean and standard deviation of global R2 across all able-bodied subjects and contraction levels.

Fig. 2. The R2 versus k plot for each hand gesture. The results were mean and standard deviation of global R2 across all able-bodied subjects and contraction levels.

v –W N hN

minimize 1

تخضع لـ (7) hN ≥0

لتقييم أي من أوجه التآزر الخاصة بالمهمة التي أعادت بناء عينات الاختبار بشكل أفضل، استخدمنا تشابه جيب التمام لقياس مدى جودة إعادة البناء بين ناقل MAP المعاد بناؤه vN وناقل MAP الأصلي v لبيانات الاختبار

SN = vN · v vN · |v| (8)

Task-specific synergies that yielded the highest goodness of reconstruction were picked out and the corresponding move- ment was determined as the unknown movement for testing data. Hereafter, the proposed classification method based on muscle synergy analysis will be abbreviated as MSC.

H. Offline Performance Evaluation

H. Offline Performance Evaluation

من أجل تقييم تأثير مستويات القوة المتغيرة على تعميم خطة التصنيف المقترحة، تم تصميم ثلاثة سيناريوهات تقييم مختلفة في هذه الدراسة:

Intra-force-level evaluation, namely training and testing the (1 algorithm with sEMG data recorded under the same single force level, which is a regular practice in previ- ous studies, however, is too ideal from the real-world conditions.

Inter-force-level evaluation, namely training and testing the (2 algorithm with sEMG data recorded under different force levels, which is more approximate to the real-world conditions. 3) Multi-force-level evaluation, namely training the algo- rithm with all three force levels and testing it with single force level data. The offline evaluation was performed through a three-fold cross validation method using sEMG data collected from all three trials for each subject. Namely, data from two out of three trials were used for the estimation of task-specific synergies (or as training data for LDA classifier), and data from the remaining trial was used as testing data. We used the overall classification accuracy (OCA) to measure the classification performance, which was defined as the percentage of correctly identified movements in terms of the total number of testing movements

OCA = No. of correctly classified movements

No. of all testing movements × 100% (9)

For each subject, the algorithm was repeated three times under each force condition so that each trial of the data was fully utilized.

The performance of proposed MSC method was compared with conventional PR-based classification scheme. It has been demonstrated that feature extraction (FE) technique based on discrete Fourier transform and root (DFTR) and time- dependent

Fig. 3 illustrated the optimal number of task-specific synergies across all able-bodied subjects at three muscle contraction levels. The box plot indicated that the median of the optimal synergy number ranged from 2 to 3, which means two to three muscle synergies are sufficient to represent the corresponding sEMG data in a low dimensional space. For the convenience of analysis, the optimal synergy number was determined as three for able-bodied subjects and five for amputee subjects in view of different channel number for corresponding EMG data. **:3 الشكل**

Fig. 3 illustrated the optimal number of task-specific synergies across all able-bodied subjects at three muscle contraction levels. The box plot indicated that the median of the optimal synergy number ranged from 2 to 3, which means two to three muscle synergies are sufficient to represent the corresponding sEMG data in a low dimensional space. For the convenience of analysis, the optimal synergy number was determined as three for able-bodied subjects and five for amputee subjects in view of different channel number for corresponding EMG data.

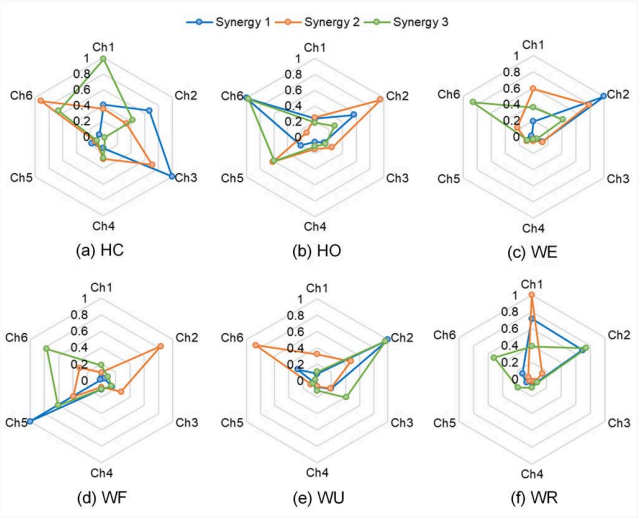


Fig. 4. Representative muscle synergies for each hand gesture performed by one able-bodied subject at medium contraction level. Each color of the radar plot represented one out of three task-specific synergies. The values were normalized to the maximum for each synergy. **:4 الشكل**

Fig. 4. Representative muscle synergies for each hand gesture performed by one able-bodied subject at medium contraction level. Each color of the radar plot represented one out of three task-specific synergies. The values were normalized to the maximum for each synergy.

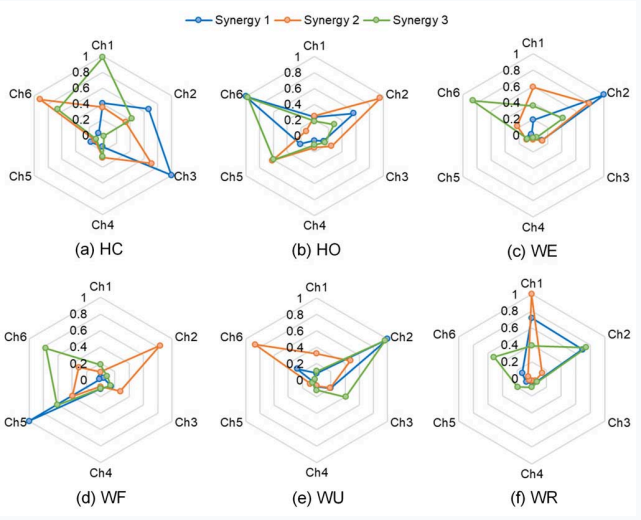


Fig. 3. Box plot for the optimal number of task-specific synergies at three different muscle contraction levels. On each box, the central mark indicates the median, and the bottom and top edges of the box indicate the 25th and 75th percentiles, respectively. The whiskers extend to the most extreme data points. The results were derived from all able-bodied subjects. **:3 الشكل**

Fig. 3. Box plot for the optimal number of task-specific synergies at three different muscle contraction levels. On each box, the central mark indicates the median, and the bottom and top edges of the box indicate the 25th and 75th percentiles, respectively. The whiskers extend to the most extreme data points. The results were derived from all able-bodied subjects.

classification accuracy. The significance level was set to 0.05 for all tests. The post-hoc analysis with Bonferroni correction method for multiple comparisons was performed when significant difference was detected.

III. RESULTS

III. RESULTS

A. Robustness of Task-Specific Synergies

A. Robustness of Task-Specific Synergies

Variation of Synergy Number: The mean and standard (1 deviation of R2 across all able-bodied subjects as functions of synergy number for each hand gesture were shown in Fig. 2. Overall, the global R2 showed an upward trend with the synergy number increasing from one to six. More specifically, a significant improvement on data reconstruction was observed with a certain synergy number (the optimal synergy number) for each movement, but after that, any further increase in the synergy number only yielded limited improvement on the value of R2.

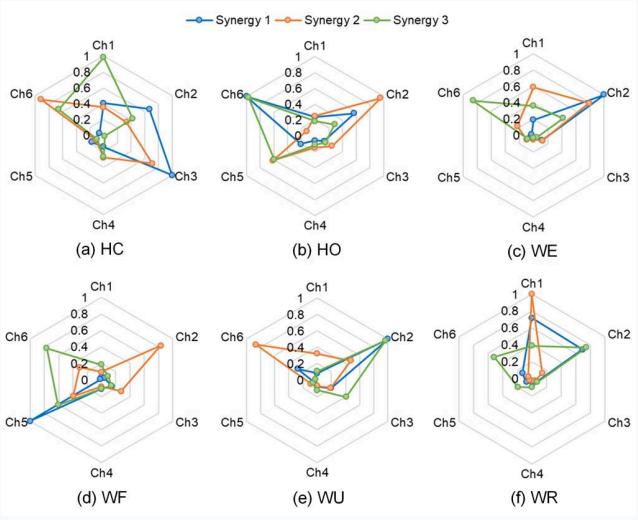


Fig. 6. Box plot for quantitative evaluation of synergy **الشكل 6:** similarity for each movement under three force variation conditions. .The results were derived from all able-bodied subjects

Fig. 6. Box plot for quantitative evaluation of synergy similarity for each movement under three force variation conditions. The results were derived from all able-bodied subjects.

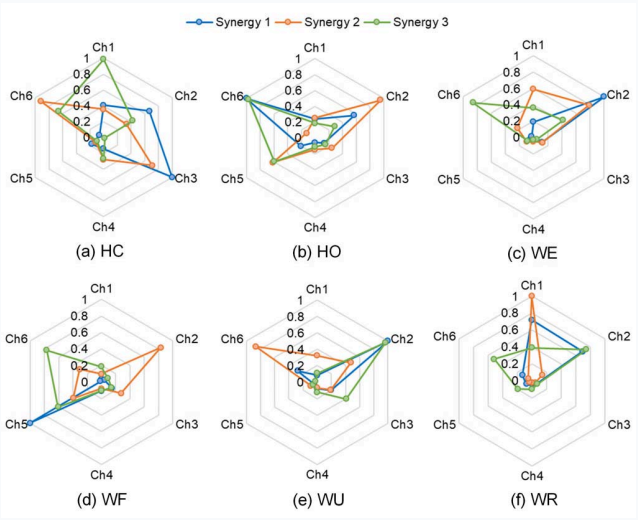


Fig. 7. Box plot for intra-movement predictive power of **الشكل 7:** task-specific synergies for each movement under nine force variation conditions. Each box represents the VAF calculated by using muscle synergies extracted at one force level to reconstruct sEMG data at another force level within the same movement. The results were .derived from all able-bodied subjects

Fig. 7. Box plot for intra-movement predictive power of task-specific synergies for each movement under nine force variation conditions. Each box represents the VAF calculated by using muscle synergies extracted at one force level to reconstruct sEMG data at another force level within the same movement. The results were derived from all able-bodied subjects.

the muscle synergies with small force variation (L-M and M-H) showed high similarity (more than 0.8) for all prede- fined hand gestures. In contrast, low similarity was observed between muscle synergies with large force variation (L-H), especially for muscle synergies associated with HC movement (0.68). Overall, the average similarity across all hand gestures and all force .levels was 0.85, higher than the threshold

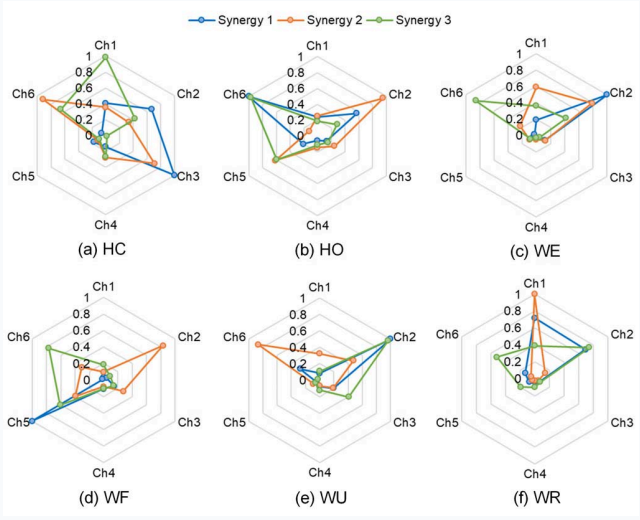


Fig. 5. A set of three task-specific synergies for WR **الشكل 5:** movement performed by one able-bodied subject. Each color represented a muscle contraction level. The values were normalized to .the maximum for each synergy

Fig. 5. A set of three task-specific synergies for WR movement performed by one able-bodied subject. Each color represented a muscle contraction level. The values were normalized to the maximum for each synergy.

performed by one able-bodied subject at medium contraction level. A set of three muscle synergies were extracted from the corresponding MAP matrix for each movement. The scaling of each axis in the radar plot represented the weighting of corresponding muscles under the specific EMG channel while performing a certain movement. The weighting values were normalized to the maximum for each synergy. It can be observed that the patterns for each set of task-specific synergies were distinct from one movement to another. The similar structural differences held true for all task-specific syn- ergies extracted .from other subjects at other muscle contraction levels

The set of three task-specific synergies associated with WR movement performed by one able-bodied subject in the setting of three muscle contraction levels were shown in Fig. 5. It can be observed that though the weighting amplitude of each muscle synergy scaled with the increase of muscle contraction level, they were quite similar in terms of the weighting balance (the global patterns in the radar chart). It meant that the structure of each synergy almost remained stable regardless of variant force levels. It was still the case for muscle synergies extracted from .other movements performed by other subjects

Fig. 8. Box plot for inter-movement predictive power of task-specific synergies under nine force variation conditions. Each box represents the VAF calculated by using muscle synergies extracted from one movement at one force level to reconstruct sEMG data from a different movement at another force level. The results were derived from all able-bodied subjects. :**الشكل 8**

Fig. 8. Box plot for inter-movement predictive power of task-specific synergies under nine force variation conditions. Each box represents the VAF calculated by using muscle synergies extracted from one movement at one force level to reconstruct sEMG data from a different movement at another force level. The results were derived from all able-bodied subjects.

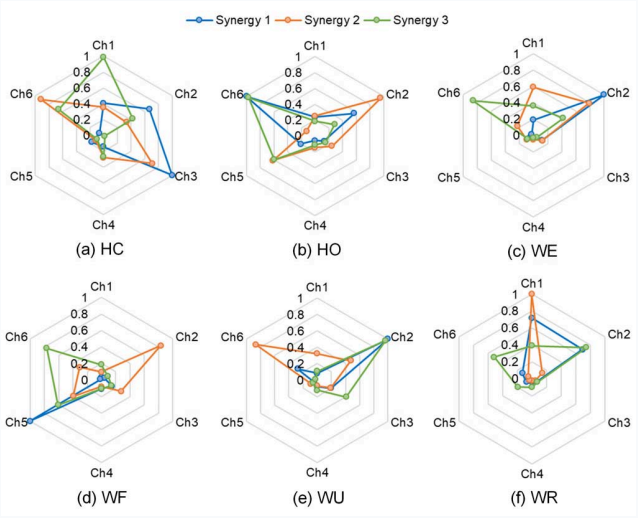


Fig. 9. Box plot for performance evaluation of six classification schemes under nine force variation conditions. (a) Training and testing with the same single force level. (b) Training and testing with different force levels. (c) Training with all three force levels and testing with single force level. The results were derived from all able-bodied subjects. :**الشكل 9**

Fig. 9. Box plot for performance evaluation of six classification schemes under nine force variation conditions. (a) Training and testing with the same single force level. (b) Training and testing with different force levels. (c) Training with all three force levels and testing with single force level. The results were derived from all able-bodied subjects.

data from other movements under nine force variation conditions. The average VAF across all force variation conditions was 0.74, 0.59, 0.67, 0.71, and 0.62 for each inter-movement combination. Apparently, there occurred a significant drop in the predictive power for inter-movement data when compared to the intra-movement results shown in Fig. 7. Such deterio- ration in the predictive power held true for muscle synergies associated with other predefined movements. These results supported the hypothesis proposed in Section II-G that task- specific synergies tended to better reconstruct the sEMG data within the same movement than those from other different movements.

C. Offline Evaluation of Synergy-Based Algorithm

C. Offline Evaluation of Synergy-Based Algorithm

Experiments on Able-Bodied Dataset: The performance of (1 six classification schemes was first evaluated on the able-bodied

B. Predictive Power of Task-Specific Synergies

B. Predictive Power of Task-Specific Synergies

Fig. 7 illustrated the predictive power of task-specific synergies for the reconstruction of intra-movement data under nine force variation conditions. Good predictive power was observed for muscle synergies associated with each predefined movement, the average VAF across all force variation condi- tions was 0.92, 0.97, 0.98, 0.95, 0.97, and 0.98, respectively. Looking into details, task-specific synergies showed better predictive power under intra-force-level conditions (with an average VAF of 0.98 across all movements for L-L, M-M, and H-H) than inter-force-level conditions (with an average VAF of 0.96 across all movements for L-M, L-H, M-L, M-H, H-L, and H-M). The predictive power continued to fall for intra-movement data with large force variations. For example, the average VAF declined to 0.92 and 0.95 for L-H and H-L conditions. Moreover, the muscle synergies extracted under medium and high force level showed better predictive power (with an average VAF of 0.98 and 0.97) for the reconstruction of data under other force levels. In contrast, a slight decline in predictive power (with an average VAF of 0.95) was observed for muscle synergies extracted at low force level. :**الشكل 7**

Fig. 7 illustrated the predictive power of task-specific syn- ergies for the reconstruction of intra-movement data under nine force variation conditions. Good predictive power was observed for muscle synergies associated with each predefined movement, the average VAF across all force variation condi- tions was 0.92, 0.97, 0.98, 0.95, 0.97, and 0.98, respectively. Looking into details, task-specific synergies showed better predictive power under intra-force-level conditions (with an average VAF of 0.98 across all movements for L-L, M-M, and H-H) than inter-force-level conditions (with an average VAF of 0.96 across all movements for L-M, L-H, M-L, M-H, H-L, and H-M). The predictive power continued to fall for intra-movement data with large force variations. For example, the average VAF declined to 0.92 and 0.95 for L-H and H-L conditions. Moreover, the muscle synergies extracted under medium and high force level showed better predictive power (with an average VAF of 0.98 and 0.97) for the reconstruction of data under other force levels. In contrast, a slight decline in predictive power (with an average VAF of 0.95) was observed for muscle synergies extracted at low force level.

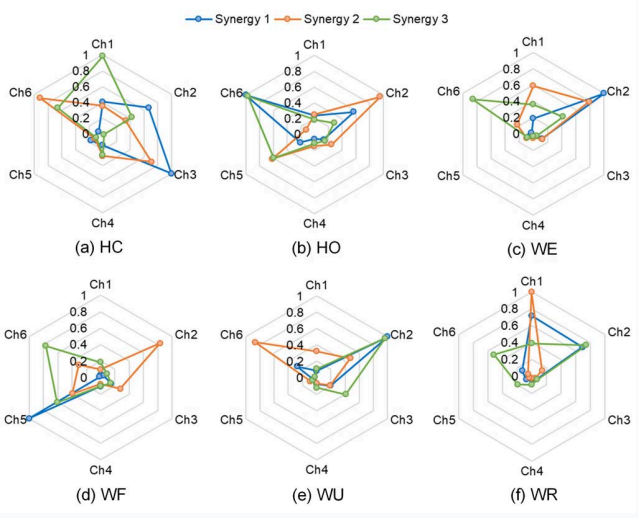


Fig. 8 illustrated the predictive power of task-specific synergies associated with HC movement for the reconstruction of :**الشكل 8**

Fig. 8 illustrated the predictive power of task-specific syn- ergies associated with HC movement for the reconstruction of

Experiments on Amputee Dataset: We further evaluated the (2 performance of different classification schemes on the amputee dataset. Fig. 10 depicted the classification accuracy achieved under nine force variation conditions for MSC + DFTR, LDA + DFTR and LDA + TDPSD, as the superiority of these methods have been demonstrated in Fig. 9 and in previous literature. Obviously, the performance on amputee subjects was not as good as that on able-bodied subjects for all classification schemes investigated. Specifically, the

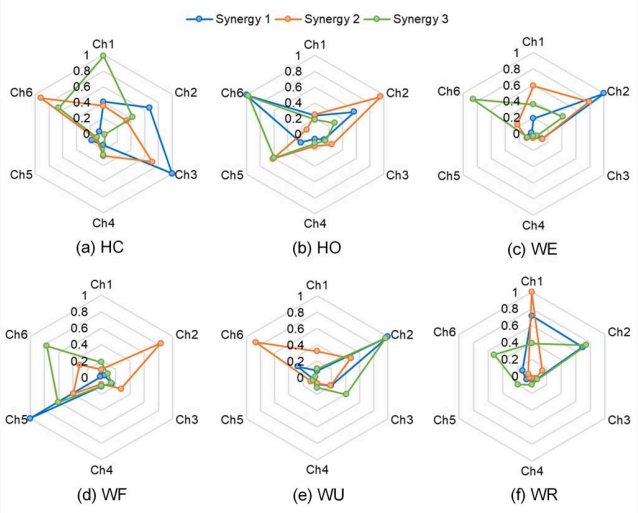


Fig. 10. Box plot for performance evaluation of representative classifi- cation schemes under nine force variation conditions. The results were derived from all amputee subjects. **الشكل 10:**

Fig. 10. Box plot for performance evaluation of representative classifi- cation schemes under nine force variation conditions. The results were derived from all amputee subjects.

TABLE I COMPUTATIONAL EFFICIENCY FOR SYNERGY-BASED CLASSIFICATION

SCHEME WITH REGARD TO DIFFERENT FEATURE

EXTRACTION TECHNIQUES

average accuracy of MSC + DFTR dropped by 13.1% and 13.9% for intra-force-level scenario (L-L, M-M, and H-H) and multi-force-level scenario (A-L, A-M, and A-H) respec- tively. As for the inter-force-level scenario (L-MH, M-LH, and H-LM), the average accuracy dropped even further by 17.4%. Statistical tests showed significant differences between MSC + DFTR and other classification schemes in inter-force- level scenario, with $p < 0.001$ for MSC + DFTR versus LDA + DFTR, $p = 0.0157$ for MSC + DFTR versus LDA + TDPSD. No significant difference was observed in intra-force-level scenario ($p = 0.619$) and multi-force-level scenario ($p = 0.733$). Overall, the average accuracy across all nine force variation conditions were 71.2%, 70.5%, and 71.5% for each classification scheme.

Computational Efficiency: Finally, we investigated the (3 computational efficiency of proposed MSC method to eval- uate dataset. Fig. 9 illustrated the classification accuracy achieved in three evaluation scenarios, with an average overall accuracy of 85.2%, 85.3%, 85.8%, 82.5%, 85.6%, and 83.4% for MSC + RMS, MSC + MAV, MSC + DFTR, LDA + TD, LDA + DFTR, and LDA + TDPSD, respectively. In intra-force-level scenario, all classification schemes showed high accuracy when training and testing with sEMG data acquired within the same single force level. The average accuracy in this case (L-L, M-M, and H-H) were 90.2%, 89.4%, 91.3%, 89.5%, 91.4%, and 91.2% for each classification scheme. No significant difference was observed between the performance of six classification schemes ($p = 0.353$).

When it comes to inter-force-level evaluation scenario, namely training and testing with sEMG data collected at different force levels, the accuracy saw a significant decline for all classification schemes. The average accuracy in this case (L-MH, M-LH, and H-LM) were 78.1%, 78.7%, 78.1%, 68.4%, 76.9%, and 71.2% for each classification scheme. Statistical tests revealed significant difference in accuracy between six classification schemes ($p = 0.001$). The pair- wise post-hoc analysis showed that the differences between three synergy-based classification schemes were statistically insignificant ($p > 0.05$). However, the performance of synergy-based method, such as MSC + DFTR, was signifi- cantly higher than LDA + TD scheme (mean difference: 9.7%, $p = 0.027$) and LDA + TDPSD scheme (mean difference: 6.9%, $p = 0.041$). It is worth noting that the synergy-based method achieved higher accuracy when training at medium/high force level than that training at low force level. The average accuracy of MSC + DFTR when training at low, medium and high force level was 82.4%, 87.9%, and 85.8%, respectively. This result was in line with the outcome of the predictive power of task-specific synergies in Section III-B. Statistical test revealed significant effects of training strategy on the generalization accuracy of MSC + DFTR method ($p = 0.027$). A pair-wise post-hoc analysis indicated the highest accuracy was achieved when training at medium force level (medium-low difference: 5.5%, $p = 0.033$). No significant difference was observed when training at medium and high force level ($p > 0.05$).

In multi-force-level evaluation scenario, better performance was observed for all classification schemes as compared to inter-force-level scenario, but the overall accuracy in this case was not as high as that in intra-force-level scenario. The average accuracy across three force variation conditions (A-L, A-M, A-H) were 87.3%, 87.7%, 88.2%, 86.9%, 88.6%, and 88.1% for each classification scheme. Statistical tests showed no significant difference between the performance of six classification schemes in this case ($p = 0.695$).

utilized a post-processing procedure to improve the accuracy and their experimental scheme was constrained within the same force level, which is too ideal for the real-world conditions. In contrast, we extended the constraints into variant force levels and employed no post- processing procedure. Besides, we chose the NMF/ANLS algorithm and NNLS algorithm to estimate muscle syner- gies and neural commands respectively, which is more cost- effective and the results are of greater physical importance. To our best knowledge, this was the first time to concurrently demonstrate the robustness and predictive power of task- specific synergies under variant force levels and explore their application in reliable movement classification against force variation.

We compared the performance of synergy-based method with conventional PR-based method consisting of LDA clas- sifier and representative FE techniques reported in literature. The classification accuracy saw a significant decline for all classification schemes when sEMG data used for training and testing were from different force levels (Fig. 9). This result agrees with the results of previous studies that force variation may degrade the performance of myoelectric classification [3]. Statistical tests indicated better performance of proposed method against force variation than conventional PR-based method. This may suggest potential application of synergy-based classification method into clinically viable prosthetics against force variation.

The proposed method showed relatively higher accuracy when trained at medium and high force levels than trained at low force level (Fig. 9). It means that reasonable train- ing strategy with data acquired at a single force level (medium or high) can significantly enhance the generaliza- tion of MSC method in the setting of variant force levels. Although the accuracy improvement in this case was less than that training with multiple force levels (Fig. 9), this synergy-based classification scheme may largely reduce the training time for potential application in upper-limb prosthe- ses with intuitive and robust myoelectric control. Moreover, the processing time required for the estimation of neural commands and movement classification is approximately 1.364 ms, which shows good prospect for real time prosthesis control.

The limitation of this study lies within two aspects: Firstly, the present method was focused on steady-state portion of EMG data with limited transient portion between tasks. How- ever, this might not be a significant problem since the effect of transient information on the classification performance could be minimized by using a threshold to detect the onset of the motion [44]. Secondly, the proposed method was only evaluated through offline analysis since the main objective of this study was to check the utility of synergy-based method against force its potential for clinically viable myoelectric control. The experiment was performed using MATLAB 2016b on a computer with 3.6 GHz Intel Core i7 processor, 16GB RAM. TABLE I listed the average computational efficiency associ- ated with feature extraction (FE) and gesture classification (GC), from overlapped analysis window of 200 ms length and 25 ms increment. The time required for gesture classifica- tion (GC) is 1.364 ms for MSC method. The controller delay for proposed scheme is therefore within 120 ms according to [42], which is applicable for myoelectric controlled pros- thesis [43].

IV. DISCUSSION

IV. DISCUSSION

Previous studies either independently demonstrated the exis- tence of MS in human motor control, or simply integrated MS into movement classification under equal muscle contraction level. However, no prior study has concurrently investigated the effect of force variation on both the synergy structure and synergy-based classification performance. The main strength of present study was that it gave an experimental insight into the reliability of MS in the setting of variant force levels. We also proposed a synergy-based classification method against force variation, which is a serious handicap in real- world myoelectric prostheses control. The proposed method is very different from conventional PR-based method that uses different features to train a classifier for task identification. Instead, we extract task-specific synergies from multichannel EMG and utilize their predictive power for task identification. Ultimately, we want to provide an alternative approach for the development of clinically viable prostheses and rehabilitation robots powered by sEMG.

We first investigated the underlying physiological informa- tion hidden in acquired sEMG signals by means of muscle synergy analysis. Task-specific synergies extracted from six predefined hand gestures showed significant structural differ- ences (Fig. 4) due to synergy specificity previously described in [39]. In contrast, high similarity (scaling amplitude with almost fix weighting balance) was observed among synergies extracted within the same movement in the presence of variant force levels (Fig. 5). This is in line with the results of [32] that muscle synergies are relatively fixed across variant con- straints. Large force variation was more likely to change the synergy structure since the similarity between L-H force level were lower than the similarity between L-M force level and M-H force level (Fig. 6). In addition, task-specific synergies showed better predictive power for intra-movement data than for inter-movement data (Fig. 7 and Fig. 8). These outcomes are the cornerstone for the proposed movement classification method. We noted that a similar classification method was introduced in [16] and the reported performance was better than ours. However, they

variation. We thus suppose that offline analysis is sufficient to reach the goal at the moment. In fact, the construction of an online system for the control of a prosthetic hand is underway and the detailed results of these trials will be reported in future .publications

V. CONCLUSION

V. CONCLUSION

In this paper, we investigated the robustness and predictive power of task-specific synergies under three self-defined muscle contraction levels, and presented a reliable classification scheme against force variation. Muscle synergy analysis of 6-channel sEMG signals showed that a three-synergy model associated with each of six predefined hand gestures was able to explain at least 90% of the variance in corresponding MAP matrix; The weighting balance of each task-specific synergies remained stable regardless of force variation, though their weighting amplitude scaled with the increase of muscle con- traction level. These results support muscle synergy analysis as a robust tool to interpret the neurophysiological information and motor intent hidden in sEMG signals; Statistical tests indicated better performance of proposed method against force variation than conventional PR-based method. We recom- mended a reasonable training strategy for satisfactory accuracy by training the scheme at medium or high force level. The controller delay of MSC method lies within 120 ms owing to the NMF/ANLS algorithm and NNLS algorithm used in the estimation of muscle synergies and neural commands respectively. These outcomes throw light on our future work towards clinically reliable myoelectric control of upper-limb prostheses. It also provides an alternative method for the development of more intuitive and natural human machine interfaces in rehabilitation robots and so .on

REFERENCES

REFERENCES

[1] D. Sharma et al., "Shadow hand," J. Advance Res. Appl. Sci., vol. 1, no. 1, pp. 4–7, 2014. [2] O. W. Samuel et al., "Intelligent EMG -pattern recognition control method for upper-limb multifunctional prostheses: Advances, current challenges, and future prospects," IEEE Access, vol. 7, pp. 10150–10165, 2019. [3] E. Campbell, A. Phinyomark, -and E. Scheme, "Current trends and confounding factors in myoelectric control: Limb position and contraction intensity," Sensors, vol. 20, no. 6, p. 1613, Mar. 2020. [4] Coapt LLC. (2017). Complete Control (Coapt LLC). [Online]. :Available https://coaptengineering.com [5] E. Biddiss and T. Chau, "Upper-limb prosthetics: Critical factors in device abandonment," Amer. J. Phys. Med. Rehabil., vol. 86, no. 12, pp. 977–987, 2007. [6] J. Davidson, "A survey of the satisfaction of upper limb amputees with their prostheses, their lifestyles, and their abilities," J. Hand Therapy, vol. 15, no. 1, pp. 62–70, 2002. [7] B. Tolooshams and -N. Jiang, "Robustness of frequency division technique for online myoelectric pattern recognition against contraction-level variation," Frontiers Bioeng. Biotechnol., vol. 5, pp. 1–9, Feb. 2017. [8] A. Pradhan, N. Jiang, V. Chester, and U. -Kuruganti, "Linear regression with frequency division technique for robust simultaneous and proportional myoelectric control during medium and high contraction- level variation," Biomed. Signal Process. Control, vol. 61, Aug. 2020, Art. no. 101984. [9] X. Li et al., "A new approach to mitigate the effect of force variation on pattern recognition for myoelectric control," in Proc. 38th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), Aug. 2016, pp. 1684–1687. [10] A. Moin, A. Zhou, S. Benatti, A. Rahimi, L. ,Benini, and J. M. Rabaey "Analysis of contraction effort level in EMG-based gesture recognition using hyperdimensional computing," in Proc. IEEE Biomed. Circuits Syst. Conf. (BioCAS), Oct. 2019, pp. 1–4. [11] O. S. Powar and K. Chemmangat, "Dynamic time warping for reducing the effect of force variation on myoelectric control of hand prostheses," J. Electromyogr. Kinesiol., vol. 48, pp. 152–160, Oct. 2019. [12] A. Al-Timemy, R. Khushaba, G. Bugmann, and J. Escudero, "Improving the performance against force variation of EMG controlled multifunc- tional upper-limb prostheses for transradial amputees," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 24, no. 6, pp. 650–661, Jun. 2016. [13] J. He, D. Zhang, X. Sheng, S. Li, and X. Zhu, "Invariant surface EMG feature against varying contraction level for myoelectric control based on muscle coordination," IEEE J. Biomed. Health Inform., vol. 19, no. 3, pp. 874–882, May 2015. [14] N. V. Iqbal, K. Subramaniam, and S. A. P., "Robust feature sets for contraction level invariant control of upper limb myoelectric prosthesis," Biomed. Signal Process. Control, vol. 51, pp. 90–96, May 2019. [15] F. Onay and A. Mert, "Phasor represented EMG feature extraction against varying contraction level of prosthetic control," Biomed. Signal Process. Control, vol. 59, May 2020, Art. no. 101881. [16] G. Rasool, K. Iqbal, N. Bouaynaya, and G. White, "Real-time task

framework for the EMG patterns of new hand postures," J. Neural Eng., vol. 6, no. 3, Jun. 2009, Art. no. 036004. [28] S. Zhang, X. Zhang, S. Cao, X. Gao, X. Chen, and P. Zhou, "Myoelectric

pattern recognition based on muscle synergies for simultaneous control of dexterous finger movements," IEEE Trans. Human-Mach. Syst., vol. 47, no. 4, pp. 576–582, Aug. 2017. [29] G. K. Patel, C. Castellini, J. M. Hahne, D. Farina, and S. ,Dosen

A classification method for myoelectric control of hand" prostheses inspired by muscle coordination," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 26, no. 9, pp. 1745–1755, Sep. 2018. [30] J. Taborri, E. Palermo, Z. D. Prete, and S. Rossi, "On the reliability

and repeatability of surface electromyography factorization by muscle synergies in daily life activities," Appl. Bionics Biomech., vol. 2018, pp. 1–15, Nov. 2018. [31] H. Deng, V. C. K. Cheung, Y. ,Geng, M. G. A. Samuel, O. W. Samuel

and G. Li, "Robustness of muscle synergies under variant muscle contraction force during forearm movements," in Proc. 42nd Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), Montreal, QC, Canada, Jul. 2020, pp. 3306–3309. [32] Y. Geng, H. Deng, O. ,W. Samuel, V. Cheung, L. Xu, and G. Li

Modulation of muscle synergies for multiple forearm" movements under variant force and arm position constraints," J. Neural Eng., vol. 17, no. 2, Apr. 2020, Art. no. 026015. [33] R. N. ,Khushaba, A. Al-Timemy, S. Kodagoda, and K. Nazarpour

Combined influence of forearm orientation and muscular" contraction on EMG pattern recognition," Expert Syst. Appl., vol. 61, pp. 154–161, Nov. 2016. [34] A. Saito, A. Tomita, R. Ando, K. Watanabe, and H. Akima, "Muscle

synergies are consistent across level and uphill treadmill running," Sci. Rep., vol. 8, no. 1, p. 5979, Apr. 2018. [35] C. B. Hart -and S. F. Giszter, "Distinguishing synchronous and time

varying synergies using point process interval statistics: Motor primitives in frog and rat," Frontiers Comput. Neurosci., vol. 7, no. 52, pp. 1–14, 2013. [36] L. H. Ting et al., "Neuromechanical principles underlying movement

modularity and their implications for rehabilitation," Neuron, vol. 86, no. 1, pp. 38–54, Apr. 2015. [37] H. Kim and H. Park, -"Nonnegative matrix factorization based on alter

nating nonnegativity constrained least squares and active set method," SIAM J. Matrix Anal. Appl., vol. 30, no. 2, pp. 713–730, 2008. [38] A. d'Avella, A. Portone, L. Fernandez, and F. Lacquaniti, "Control of

discrimination for myoelectric control employing task-specific muscle synergies," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 24, no. 1, pp. 98–108, Jan. 2016. [17] M. C. Tresch, P. Saltiel, and E. Bizzi, "The construction of movement

by the spinal cord," Nature Neurosci., vol. 2, pp. 162–167, Feb. 1999. [18] N. Jiang, J. L. Vest-Nielsen, S. Muceli, and D. Farina, "EMG-based

simultaneous and proportional estimation of wrist/hand kinematics in uni-lateral trans-radial amputees," J. Neuroeng. Rehabil., vol. 9, no. 1, p. 42, 2012. [19] N. Jiang, H. Rehbaum, I. ,Vujaklija, B. Graimann, and D. Farina

Intuitive, online, simultaneous, and proportional myoelectric" control over two degrees-of-freedom in upper limb amputees," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 22, no. 3, pp. 501–510, May 2014. [20] J. Ma, N. V. Thakor, and F. Matsuno, "Hand and wrist movement control

of myoelectric prosthesis based on synergy," IEEE Trans. Human-Mach. Syst., vol. 45, no. 1, pp. 74–83, Feb. 2015. [21] M. -Ison, I. Vujaklija, B. Whitsell, D. Farina, and P. Artemiadis, "High

density electromyography and motor skill learning for robust long-term control of a 7-DoF robot arm," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 24, no. 4, pp. 424–433, Apr. 2016. [22] M. Ison, I. Vujaklija, B. Whitsell, D. Farina, and P. Artemiadis, -"Simulta

neous myoelectric control of a robot arm using muscle synergy-inspired inputs from high-density electrode grids," in Proc. IEEE Int. Conf. Robot. Automat. (ICRA), Seattle, WA, USA, .May 2015, pp. 6469–6474

[23] G. R. Naik and H. T. Nguyen, "Nonnegative matrix factorization for

the identification of EMG finger movements: Evaluation using matrix analysis," IEEE J. Biomed. Health Inform., vol. 19, no. 2, pp. 478–485, Mar. 2015. [24] C. W. Antuvan, F. Bisio, F. Marini, S.-C. Yen, E. Cambria, and

L. Masia, "Role of muscle synergies in real-time classification of upper

L. Masia, "Role of muscle synergies in real-time classification of upper

limb motions using extreme learning machines," J. Neuroeng. Rehabil., vol. 13, no. 1, p. 76, Dec. 2016. [25] X. Luo et al., -"Synergistic myoelectrical activities of forearm mus

cles improving robust recognition of multi-fingered gestures," Sensors, vol. 19, no. 3, p. 610, Feb. 2019. [26] B. Atoufi, E. N. -Kamavuako, B. Hudgins, and K. Englehart, "Classifi

cation of hand and wrist tasks of unknown force levels using muscle synergies," in Proc. 37th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), Milan, Italy, Aug. 2015, pp. 1663–1666. [27] A. B. Ajiboye and R. F. Weir, "Muscle synergies as a predictive

fast-reaching movements by muscle synergy combinations," J. Neurosci., vol. 26, no. 30, pp. 7791–7810, 2006. [39] A. d'Avella and E. Bizzi, "Shared and specific muscle synergies

in natural motor behaviors," Proc. Nat. Acad. Sci. USA, vol. 102, pp. 3076–3081, Feb. 2005. [40] R. Bro and S. D. Jong, "A fast non-negativity-constrained least squares

algorithm," J. Chemometrics, vol. 11, no. 5, pp. 393–401, 1997. [41] B. Hudgins, P. Parker, and R. N. Scott, "A new strategy for -multifunc

tion myoelectric control," IEEE Trans. Biomed. Eng., vol. 40, no. 1, pp. 82–94, Jan. 2002. [42] T. R. Farrell, "Determining delay

created by multifunctional prosthesis

controllers," J. Rehabil. Res. Develop., vol. 48, no. 6, 2011. [43] T. R. Farrell and R. F. Weir, "The optimal controller delay for -myoelec

tric prostheses," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 15, no. 1, pp. 111–118, Mar. 2007. [44] T. Lorrain, N. Jiang, and D. Farina, "Influence of the training set on

the accuracy of surface EMG classification in dynamic contractions for the control of multifunction prostheses," J. .Neuroeng. Rehabil., vol. 8, p. 25, May 2011